

# Bacterial Phosphatidylethanolamine Biosynthesis Through Phosphatidylserine: A Two-Reaction Module (PssA → Psd)

---

*A commissioned review-style synthesis for a molecular biology audience*

---

## 1. Executive Summary

---

Most bacteria manufacture phosphatidylethanolamine (PE) — often the single most abundant membrane phospholipid (~70–80% in *Escherichia coli*) — through a compact, two-reaction module that branches off the central lipid intermediate CDP-diacylglycerol (CDP-DAG). In the first, committed step, **phosphatidylserine synthase (PssA)** transfers the phosphatidyl group of CDP-DAG onto L-serine, releasing CMP and producing **phosphatidylserine (PS)**. In the second step, **phosphatidylserine decarboxylase (Psd)** removes the carboxyl group of the serine headgroup to yield PE and CO<sub>2</sub>. The two enzymes are functionally coupled: PssA supplies the PS substrate that Psd consumes, and in the canonical route PS is essentially a transient intermediate that does not accumulate.

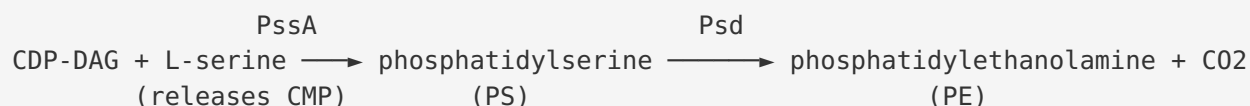
The most striking feature of this system is an **asymmetry between its two halves**. The first enzyme, PssA, is not a single conserved protein but two mechanistically and structurally unrelated protein families that have *convergently* evolved to perform the same phosphatidyltransfer reaction: **Type I** peripheral/amphitropic enzymes of the phospholipase-D (PLD/HKD) superfamily (exemplified by *E. coli* PssA, which forms a covalent phosphatidyl-enzyme intermediate), and **Type II** integral-membrane enzymes of the CDP-alcohol phosphatidyltransferase (CDP-AP) superfamily (exemplified by *Bacillus subtilis* and *Sinorhizobium meliloti*). By contrast, the second enzyme, Psd, is a remarkably *uniform* and deeply conserved self-cleaving, **pyruvoyl-dependent decarboxylase** whose catalytic chemistry — autoprocessing of a proenzyme into  $\alpha/\beta$  subunits followed by Schiff-base-mediated decarboxylation — is shared from bacteria to human mitochondria.

Finally, the **necessity of this module is lineage-dependent** and should not be over-generalized from *E. coli*. In *E. coli*, loss of PssA or Psd abolishes PE, imposes an absolute requirement for divalent cations, and disrupts cytokinesis. In other bacteria, PE is dispensable under some conditions (*Sinorhizobium meliloti*) or can be made by Psd-independent bypass routes (*Xanthomonas campestris* bifunctional cardiolipin/PE synthase, which condenses CDP-DAG directly with ethanolamine). This review defines the boundaries of the system, lays out the best current mechanistic model, catalogs the molecular players, and maps evolutionary and physiological variation — while being explicit about where evidence is strong and where it is indirect.

## 2. Definition and Biological Boundaries

### 2.1 What is included

The system reviewed here is the **two-reaction bacterial PE biosynthetic module** that begins at CDP-DAG and ends at PE:



- **Step 1 — PS formation (PssA):** a CDP-diacylglycerol–serine *O*-phosphatidyltransferase reaction (EC 2.7.8.8). CDP-DAG is the phosphatidyl donor; L-serine is the acceptor; CMP is released.
- **Step 2 — PE formation (Psd):** a phosphatidylserine decarboxylase reaction (EC 4.1.1.65). The  $\alpha$ -carboxyl of the serine headgroup is removed, converting the serine headgroup to ethanolamine.

The upstream boundary is CDP-DAG (the shared branch-point liponucleotide feeding several phospholipid pathways). The downstream boundary is PE. PE feeds forward into further reactions (e.g., PE  $\rightarrow$  phosphatidylcholine by successive methylation in some bacteria, or PE as an acyl/ethanolamine donor), but those are outside the module.

## 2.2 Neighboring processes that should be treated separately

1. **The Pcs (phosphatidylcholine synthase) route to PC.** In *Sinorhizobium meliloti* and other  $\alpha$ -proteobacteria, choline is condensed directly with CDP-DAG to form phosphatidylcholine, entirely bypassing PS and PE (PMID: 14996797). This is a parallel headgroup-transfer reaction to a *different* aminoalcohol and is not part of the PE module.
2. **Direct ethanolamine condensation (CL/PE synthase bypass).** A cardiolipin-synthase-family enzyme in *Xanthomonas campestris* (Xc\_0186) condenses CDP-DAG with free ethanolamine to yield PE directly, without going through PS or Psd (PMID: 24707916). This achieves the same product by different chemistry and should be treated as an alternative route.
3. **The CDP-ethanolamine (Kennedy) pathway.** In many eukaryotes and some bacteria, exogenous ethanolamine is phosphorylated, activated to CDP-ethanolamine, and transferred onto diacylglycerol — same product, unrelated chemistry and enzymes.
4. **Eukaryotic, organelle-compartmentalized PSD reactions.** Yeast and animal cells run the *same* decarboxylation chemistry but distribute it across mitochondria (Psd1) and Golgi/vacuolar compartments (Psd2), and possess PS-independent routes (PMID: 20044027). These are evolutionary descendants but involve additional trafficking and regulatory layers absent in the streamlined bacterial module.
5. **Pyruvoyl-dependent decarboxylases acting on other substrates.** The Psd catalytic scaffold has been repurposed in some organisms; e.g., in *Aspergillus oryzae* a PSD homolog functions as an **arginine decarboxylase** (ADC1) using the same pyruvoyl chemistry (PMID: 39321488). Same mechanism, different substrate — not part of PE biosynthesis.

## 2.3 Competing definitions

The literature is largely consistent on the *reactions*, but definitions diverge on the **enzyme classification of PssA**. The "Type I / Type II" nomenclature is the most useful and best-supported framing: Type I refers to the PLD-superfamily peripheral enzymes (Gammaproteobacteria such as *E. coli*), and Type II to the CDP-AP integral-membrane enzymes (*Bacillus*, *Sinorhizobium*, and many others) (PMID: 14996797, PMID: 39693441). This dual-family reality means "PssA" is a *functional* label, not a single ortholog group — an important caveat for comparative genomics. A second hazard: "phosphatidylserine synthase" in mammals denotes  $\text{Ca}^{2+}$ -dependent base-exchange enzymes (PSS1/PSS2) that are mechanistically unrelated to bacterial PssA and should not be conflated with it.

### 3. Key Findings

---

#### F001 — Bacterial PE is made by a conserved two-enzyme module: PssA then Psd

Multiple structural and genetic studies converge on a two-reaction route from CDP-DAG: PssA condenses the phosphatidyl group with L-serine to give PS (releasing CMP); Psd then decarboxylates PS to PE. PE is a major bacterial membrane lipid. The terminal step is defined by the observation that *"Phosphatidylethanolamine (PE), a major component of the cellular membrane across all domains of life, is synthesized exclusively by membrane-anchored phosphatidylserine decarboxylase (PSD) in most bacteria"* (PMID: 33707636).

The genetic logic that PssA is the committed first step is clear in *E. coli*: a *pssA* deletion strain *"lacked phosphatidylserine synthase and phosphatidylethanolamine and required divalent metal ions for growth"* (PMID: 8824831). The physiological importance of the endpoint is shown by the fact that PE-less cells (a *pss-93* null) *"are defective in cell division"* (PMID: 9696776). Together these establish an obligate, ordered relay: PssA makes PS, Psd converts PS to PE, and loss of PE has severe physiological consequences in *E. coli*.

#### F002 — PssA occurs as two mechanistically distinct families: Type I (PLD-superfamily) and Type II (CDP-alcohol phosphatidyltransferase)

Bacterial PS synthases split into two non-homologous architectures that arrived at the same catalytic outcome by independent evolution. **Type I** (e.g., *E. coli* PssA) is a peripheral/amphitropic membrane protein of the phospholipase-D (HKD) superfamily that catalyzes via a covalent phosphatidyl-enzyme intermediate. Crystallographic work shows that *"the membrane-bound enzyme acts on cytidine diphosphate diacylglycerol (CDP-DG) to form cytidine monophosphate and a covalent intermediate, which is subsequently targeted by serine to produce phosphatidylserine"* (PMID: 39693441).

**Type II** (e.g., *Bacillus subtilis*, *Sinorhizobium meliloti*) is an integral-membrane enzyme of the CDP-alcohol phosphatidyltransferase family; the *S. meliloti* enzyme *"belongs to the type II phosphatidylserine synthases"* (PMID: 14996797). A functional distinction emerged from cross-complementation of an *E. coli pssA* mutant: with the membrane-bound *Bacillus* (Type II) enzyme, *"the phosphatidylethanolamine content was dependent on its activity, in contrast to that with the soluble E. coli counterpart"* (PMID: 8824831). The two families are a classic example of **convergent evolution of enzymatic function**: same substrates, same product, unrelated folds and mechanisms.

## F003 — Psd is a self-cleaving, pyruvoyl-dependent decarboxylase using a Schiff-base mechanism

Psd is synthesized as a proenzyme that autocatalytically cleaves at a conserved (L)GS(S/T) motif into a large **β-subunit** and a small **α-subunit**; the cleavage generates an N-terminal **pyruvoyl** prosthetic group on the α-subunit: *"The covalently attached pyruvoyl moiety is formed in a concerted reaction when the PSD proenzyme undergoes an endoproteolytic cleavage into a large β-subunit, and a smaller α-subunit, which harbors the prosthetic group at its N terminus"* (PMID: 25724650).

The self-processing is serine-protease-like: sequence analysis places PSDs in a D-H-S family, *"suggesting that PSDs belong to the D-H-S serine protease family"* (PMID: 25724650), and in *E. coli* *"E. coli PSD primarily employs D90/D142-H144-S254 to achieve auto-cleavage for the proenzyme maturation"* (PMID: 33707636); PMSF (a serine-protease inhibitor) blocks processing. Once matured, *"The enzyme undergoes auto-cleavage for activation and utilizes the pyruvoyl moiety to form a Schiff base intermediate with PS to facilitate decarboxylation"* (PMID: 33707636). The active *E. coli* enzyme forms a homodimer, each protomer bearing a positively charged substrate-binding pocket, with an N-terminal hydrophobic helical region mediating monotopic membrane binding and efficient PS capture — explaining broad tolerance of different PS acyl species (PMID: 32402247).

## F004 — Alternative and bypass routes exist; the PssA/Psd module is not universally essential

PE can be produced or bypassed by non-canonical means. *Xanthomonas campestris* encodes a bifunctional cardiolipin/PE synthase (Xc\_0186) that condenses CDP-DAG with ethanolamine to make PE directly, independent of PS/Psd; the authors *"consider Xc\_0186 the founding member of a new class of enzymes called CL/PE synthase (CL/PEs)"* (PMID: 24707916). A *psd* deletion loses PE and accumulates PS, but PE synthesis is partially restored by exogenous ethanolamine via this enzyme.

In *Sinorhizobium meliloti*, *"Phosphatidylethanolamine is not essential for growth of Sinorhizobium meliloti on complex culture media,"* and PC can be made without PE because *"choline is condensed with CDP-diacylglycerol to obtain PC directly"* via the Pcs route (PMID: 14996797). Conversely, in *E. coli* the module is effectively required (*pssA/psd* nulls are PE-less, cation-dependent, and division-impaired). Essentiality is therefore lineage- and condition-dependent, not intrinsic to the module.

## F005 — The pyruvoyl-Psd step is deeply conserved bacteria→eukaryotes, while eukaryotes compartmentalize and multiply the enzymes

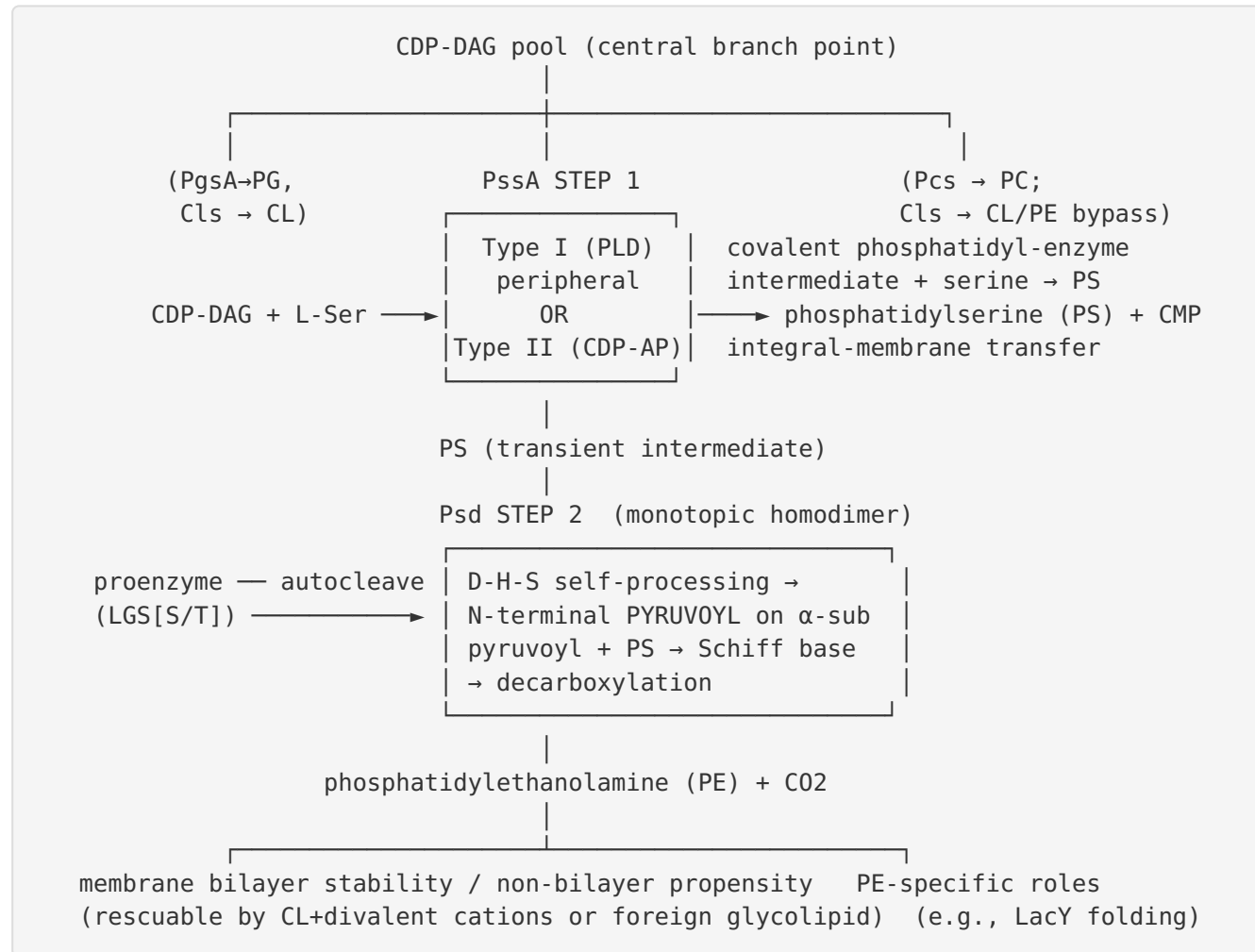
The self-cleaving pyruvoyl PSD mechanism is shared across prokaryotes and eukaryotes: "*PSDs play a central role in the synthesis of phosphatidylethanolamine in numerous species of prokaryotes and eukaryotes*" (PMID: 25724650), indicating an ancient conserved catalytic core. Eukaryotes retain the same chemistry but distribute it across organelles and gene copies: in *Saccharomyces cerevisiae*, PE is made by "*decarboxylation of phosphatidylserine (PS) by (i) phosphatidylserine decarboxylase 1 (Psd1p) in mitochondria and (ii) phosphatidylserine decarboxylase 2 (Psd2p) in a Golgi/vacuolar compartment*" (PMID: 20044027), plus two PS-independent routes; deletion of *PSD1* and/or *PSD2* depletes cellular and plasma-membrane PE. Mitochondrial Psd1 is the direct descendant of the  $\alpha$ -proteobacterial endosymbiont enzyme — a molecular fossil of the bacterial module — making streamlined bacterial Psd the best representative of the ancestral role. The upstream PS synthase, by contrast, was largely *replaced* in the animal lineage by unrelated  $\text{Ca}^{2+}$ -dependent base-exchange enzymes, illustrating that the two halves of the module have very different evolutionary trajectories.

## 4. Mechanistic Model and Interpretation

### 4.1 Best current model of the sequence of events

- 1. PssA-catalyzed phosphatidyl transfer (obligatory first step).** CDP-DAG is the activated phosphatidyl donor. In the Type I mechanism, the enzyme attacks CDP-DAG to release CMP and form a covalent phosphatidyl-enzyme intermediate; L-serine then attacks this intermediate to yield PS (PMID: 39693441). In the Type II mechanism (CDP-AP superfamily), catalysis proceeds within an integral-membrane active site, generally via a metal-assisted single-displacement without a stable covalent intermediate.
- 2. Psd self-maturation (obligatory prerequisite for the second step).** Before Psd can act, its proenzyme must autocleave at the (L)GS(S/T) motif into  $\beta$ - and  $\alpha$ -subunits, generating the N-terminal pyruvoyl group (PMID: 25724650).
- 3. Psd-catalyzed decarboxylation (obligatory second step).** The pyruvoyl carbonyl forms a Schiff base with the amino group of the PS serine headgroup, providing the electron sink that drives loss of  $\text{CO}_2$ ; hydrolysis of the resulting imine releases PE and regenerates the pyruvoyl group (PMID: 33707636, PMID: 32402247).

## 4.2 Integrated diagram



### 4.3 Obligatory, conditional, and accessory steps

| Step                                              | Status                                                         | Basis                                                                           |
|---------------------------------------------------|----------------------------------------------------------------|---------------------------------------------------------------------------------|
| PssA phosphatidyl transfer to serine              | <b>Obligatory</b> in the canonical route (only way to make PS) | <i>pssA</i> null → no PS, no PE (PMID: 8824831)                                 |
| Psd proenzyme self-cleavage → pyruvoyl            | <b>Obligatory</b> prerequisite for decarboxylation             | Cleavage generates the catalytic prosthetic group (PMID: 25724650)              |
| Psd Schiff-base decarboxylation                   | <b>Obligatory</b> terminal step                                | Structures show pyruvoyl-PS imine (PMID: 33707636)                              |
| Divalent-cation supplementation                   | <b>Conditional</b> (only in PE-less mutants)                   | Rescues viability of PE-deficient <i>E. coli</i> (PMID: 8824831, PMID: 7961817) |
| Type I PssA membrane association                  | <b>Conditional</b> (regulatory, links activity to membrane)    | Amphitropic cytosol⇌membrane equilibrium (PMID: 39693441)                       |
| Direct ethanolamine condensation (CL/PE synthase) | <b>Accessory / bypass</b> (lineage-specific)                   | Restores PE in <i>psd</i> mutants of <i>Xanthomonas</i> (PMID: 24707916)        |



## 5. Major Molecular Players and Active Assemblies

| Player                                   | Reaction                                            | Family / fold                                           | Membrane topology                             | Key mechanistic feature                                       | Representative refs                                                     |
|------------------------------------------|-----------------------------------------------------|---------------------------------------------------------|-----------------------------------------------|---------------------------------------------------------------|-------------------------------------------------------------------------|
| <b>PssA — Type I</b>                     | CDP-DAG + serine → PS + CMP                         | Phospholipase-D (HKD) superfamily                       | Peripheral / amphitropic (cytosol ⇌ membrane) | Covalent phosphatidyl-enzyme intermediate; His nucleophile    | <i>E. coli</i> (PMID: 39693441, PMID: 8824831)                          |
| <b>PssA — Type II</b>                    | CDP-DAG + serine → PS + CMP                         | CDP-alcohol phosphatidyltransferase (CAPT)              | Polytopic integral membrane                   | Metal-dependent direct transfer, no covalent intermediate     | <i>B. subtilis</i> , <i>S. meliloti</i> (PMID: 14996797, PMID: 8824831) |
| <b>Psd</b>                               | PS → PE + CO <sub>2</sub>                           | Pyruvoyl-dependent decarboxylase; D-H-S self-processing | Monotopic (N-terminal amphipathic helices)    | Autocleaved α/β heterodimer; N-terminal pyruvoyl; Schiff base | <i>E. coli</i> (PMID: 33707636, PMID: 32402247, PMID: 25724650)         |
| <b>CL/PE synthase (Xc_0186) (bypass)</b> | CDP-DAG + ethanolamine → PE (and CDP-DAG + PG → CL) | Cardiolipin synthase (PLD) family                       | Integral membrane                             | Psd-independent direct PE synthesis                           | <i>X. campestris</i> (PMID: 24707916)                                   |

**Assembly notes.** Neither core enzyme requires a large protein complex. PssA acts as a monomer whose activity is gated by reversible membrane binding (Type I) or by constitutive membrane insertion plus divalent metal (Type II). Mature Psd is an **α/β heterodimer** (products of one gene) that functions as a homodimer of such heterodimers at the membrane surface (PMID: 32402247). The functional "assembly" that matters most is enzyme-plus-membrane: both steps are interfacial reactions on lipid substrates embedded in the bilayer.

## 6. Evolutionary and Cell-Biological Variation

---

**Across bacterial lineages.** PssA architecture is the major axis of variation:  $\gamma$ -Proteobacteria such as *E. coli* use the peripheral Type I (PLD-superfamily) enzyme; many Firmicutes and  $\alpha$ -proteobacteria use the integral-membrane Type II (CAPT) enzyme — both accomplishing identical chemistry (convergence). Psd architecture is comparatively uniform across bacteria (a single monotopic pyruvoyl enzyme), making it the more conserved half of the module. Bypass capacity also varies — *Xanthomonas* carries a CL/PE synthase that makes PE from ethanolamine independently of Psd (PMID: 24707916) — as does downstream tailoring (PE→PC methylation, or direct Pcs synthesis of PC; PMID: 14996797).

**Across domains — eukaryotic compartmentalization.** The decarboxylation chemistry is deeply conserved into eukaryotes, but enzyme count and localization expand. In *S. cerevisiae*, PE is made by mitochondrial Psd1p and Golgi/vacuolar Psd2p (plus two PS-independent routes); deleting *PSD1/PSD2* depletes cellular and plasma-membrane PE (PMID: 20044027). Mitochondrial Psd1 is the direct descendant of the bacterial endosymbiont enzyme and the best representative of the ancestral role, whereas the upstream PS synthase was largely replaced in animals by  $\text{Ca}^{2+}$ -dependent base-exchange enzymes.

**Physiological states.** Because Type I PssA is amphitropic and branch-point fluxes are co-regulated, the PE:anionic-lipid ratio can be tuned to growth conditions (PMID: 8824831). PE is enriched at specific membrane locales (division septum, cell poles) and, with cardiolipin, forms lateral microdomains implicated in division and protein localization (PMID: 16925550, PMID: 9696776).

---

## 7. Constraints, Dependencies, and Failure Modes

---

**Ordering constraints.** 1. CDP-DAG must be made before either branch proceeds; PssA and PgsA compete for it. 2. **PssA before Psd:** Psd's only substrate in this module is PS; a *pssA* null therefore also abolishes PE (PMID: 8824831). 3. **Psd self-processing before catalysis:** the pyruvoyl cofactor exists only after autocleavage, so maturation strictly precedes turnover; mutations in D90/D142-H144-S254 that block cleavage abolish activity (PMID: 33707636, PMID: 25724650).

**Compartment/substrate specificity.** Both steps are interfacial and act on bilayer-embedded substrates; enzymes must reach the membrane (PssA by reversible binding; Psd by its amphipathic N-terminus). Psd's hydrophobic acyl-chain groove makes it tolerant of diverse PS species but committed to PS headgroup geometry ([PMID: 32402247](#)).

**Failure modes.** - **Loss of PE (pssA or psd null):** in *E. coli*, PE-deficient cells survive only with high divalent-cation supplementation, show increased permeability, and are defective in cytokinesis ([PMID: 8824831](#), [PMID: 7961817](#), [PMID: 9696776](#)). This reflects PE's role as a non-bilayer-prone, hydrogen-bonding lipid: the neutral, NB-prone glycolipid MGlCDAG can substitute for many PE functions ([PMID: 14688287](#)), and PE acts as a lipid "chaperone" for folding of polytopic proteins such as lactose permease ([PMID: 10212204](#)). - **Conditional dispensability:** in *S. meliloti*, PE is not essential on complex media, and PC can be supplied by the PE-independent Pcs route ([PMID: 14996797](#)) — essentiality is not intrinsic to the module.

**Evidence that rules out alternative paths.** Genetic epistasis (a *pssA* null being PE-less; *psd* nulls accumulating PS) demonstrates that, in canonical bacteria, PE flows *only* through PS unless a dedicated bypass enzyme is present ([PMID: 8824831](#), [PMID: 24707916](#)). The requirement for a pre-formed pyruvoyl cofactor rules out a PLP-based route for Psd ([PMID: 25724650](#)). Separately, PE is *not* required for lipoprotein N-acylation in *E. coli* — other phospholipids can serve as the acyl donor ([PMID: 2033085](#)) — ruling out one long-postulated obligatory PE role.

---

## 8. Evidence Base

---

| PMID                     | Title (abbrev.)                                                            | Role in this review                                                                                                        |
|--------------------------|----------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|
| <a href="#">8824831</a>  | Regulatory mechanism for balanced phospholipid synthesis in <i>E. coli</i> | PssA is committed step; <i>pssA</i> null → no PE, cation-dependent; Type I vs Type II behavior contrast (F001, F002, F004) |
| <a href="#">39693441</a> | Structural basis for membrane association and catalysis by PS synthase     | Type I covalent phosphatidyl-enzyme mechanism (F002)                                                                       |
| <a href="#">14996797</a> | PE not essential in <i>Sinorhizobium meliloti</i>                          | Type II assignment; PE dispensability; Pcs route (F002, F004)                                                              |
| <a href="#">25724650</a> | Protease to decarboxylase: PSD metamorphosis                               | Proenzyme self-cleavage; pyruvoyl; D-H-S family; broad conservation (F003, F005)                                           |
| <a href="#">33707636</a> | Structural insights into PE formation in bacterial membranes               | Auto-cleavage, Schiff base, catalytic residues, membrane binding (F001, F003)                                              |
| <a href="#">32402247</a> | Structural basis for PE biosynthesis by bacterial PSD                      | EcPsd homodimer, substrate pocket, N-terminal helix (F003)                                                                 |
| <a href="#">9696776</a>  | Cell division proteins in PE-lacking filamentous <i>E. coli</i>            | PE-less cells defective in division (F001)                                                                                 |
| <a href="#">7961817</a>  | Lipid polymorphism essential for PE-deficient <i>E. coli</i> viability     | Cation/CL rescue via lipid polymorphism (§7)                                                                               |
| <a href="#">24707916</a> | Bifunctional cardiolipin/PE synthase                                       | Psd-independent PE bypass from CDP-DAG + ethanolamine (F004)                                                               |
| <a href="#">20044027</a> | PE from four pathways to yeast plasma membrane                             | Eukaryotic Psd1/Psd2 compartmentalization (F005)                                                                           |
| <a href="#">14688287</a> | Monoglucosyldiacylglycerol substitutes for PE                              | Foreign non-bilayer lipid rescues PE loss (§7)                                                                             |
| <a href="#">10212204</a> | PE as molecular chaperone for LacY refolding                               | Evidence for PE-specific functional roles (§7, §9)                                                                         |
| <a href="#">2033085</a>  | PE not essential for lipoprotein N-acylation                               | Rules out one obligatory PE role (§7)                                                                                      |
| <a href="#">39321488</a> | Arginine decarboxylase (PSD homolog) in <i>A. oryzae</i>                   | Pyruvoyl scaffold repurposed to non-PE substrate (§2, §9)                                                                  |

| PMID                     | Title (abbrev.)                      | Role in this review                                        |
|--------------------------|--------------------------------------|------------------------------------------------------------|
| <a href="#">16925550</a> | Lipid domains in bacterial membranes | Context: PE/CL microdomains and synthase localization (§6) |

## 9. Controversies and Open Questions

**Strongly supported (high confidence):** - The two-reaction route CDP-DAG → PS (PssA) → PE (Psd) is the canonical bacterial PE pathway (F001). - PssA exists as two convergent families, Type I (PLD-superfamily) and Type II (CDP-AP) (F002). - Psd is a self-cleaving pyruvoyl decarboxylase using serine-protease-like D-H-S autoprocessing and Schiff-base decarboxylation (F003; strong structural evidence).

**Areas of disagreement, indirect evidence, or organism mixing:** 1. **Generalizing essentiality from *E. coli*.** Much of the "PE is essential" narrative rests on *E. coli*; yet PE is dispensable in *Sinorhizobium* on rich media and bypassable in *Xanthomonas*. Claims of universal essentiality should be qualified. 2. **Mechanistic detail of Type II PssA.** The covalent-intermediate mechanism is well established for Type I (*E. coli*), but catalytic details of the integral-membrane Type II enzymes are comparatively under-characterized structurally; most CAPT mechanistic inference is transferred by homology. Extrapolating a single mechanism across both families is unsafe. 3. **Cross-organism structural inference for Psd.** The Schiff-base/pyruvoyl model is robust, but studies mix data from *E. coli*, *Plasmodium*, yeast, and fungal homologs — some repurposed to *non-PE* substrates such as arginine (PMID: [39321488](#)). Care is needed not to assume identical substrate handling across all "PSD-family" proteins. 4. **What makes PE (or its physical properties) essential?** Rescue by cardiolipin+cations or a foreign glycolipid (PMID: [7961817](#), PMID: [14688287](#)) suggests it is largely the non-bilayer/charge property, but specific chaperone-like roles of PE (PMID: [10212204](#)) argue for additional PE-specific functions. The balance remains unresolved. 5. **In-vivo regulation of Type I PssA membrane association.** The amphitropic cytosol⇌membrane equilibrium is documented biochemically and structurally, but the physiological triggers and their quantitative contribution to flux control are incompletely resolved. 6. **Druggability.** PSD's unusual pyruvoyl chemistry and self-processing are attractive, mechanism-unique antimicrobial targets, but selectivity over the conserved human mitochondrial Psd1 is an unresolved practical concern.

**Most important open questions:** the phylogenetic distribution and relative antiquity of Type I vs. Type II PssA; high-resolution structures/mechanisms of Type II PssA; how flux is partitioned at CDP-DAG among PE, PG/CL, and PC branches; and how broadly distributed the Psd-independent bypass routes are.

---

## 10. Limitations and Knowledge Gaps

---

1. **Literature synthesis, not primary data.** This review synthesizes published literature (24 papers reviewed) rather than new experiments; conclusions inherit the biases of the source studies.
  2. **E. coli-centric mechanistic depth.** High-resolution structural/mechanistic detail is concentrated on *E. coli* Type I PssA and *E. coli* Psd. Type II PssA mechanisms are comparatively under-resolved.
  3. **Uneven phylogenetic sampling.** The prevalence of Type I vs. Type II PssA across bacterial phyla, and of Psd-independent bypass enzymes, has not been systematically quantified here.
  4. **Function attribution.** Distinguishing PE's generic biophysical roles from its specific molecular-ligand roles remains partly unresolved and is confounded by rescue experiments with surrogate lipids.
  5. **Cross-kingdom extrapolation.** Some cited mechanistic support comes from eukaryotic or repurposed (non-PE) homologs; care was taken to flag these, but residual extrapolation risk remains.
- 

## 11. Proposed Follow-up Experiments / Actions

---

1. **Comparative genomics survey.** Systematically classify PssA homologs (Type I PLD-superfamily vs. Type II CDP-AP) across sequenced bacterial phyla, and co-map Psd and Cls-family CL/PE-synthase presence to quantify how often bypass routes coexist.
2. **High-resolution structure of a Type II PssA.** Solve a cryo-EM/crystal structure of an integral-membrane Type II PS synthase (e.g., *Bacillus* or *Sinorhizobium*) with substrate/product to test whether it uses a covalent intermediate or a metal-assisted direct transfer.

3. **Quantify flux partitioning at CDP-DAG.** Use isotope tracing/lipidomics in *E. coli* and a Type-II-using organism to measure how carbon is split between PE, PG/CL, and PC branches under varied conditions.
  4. **Dissect Psd maturation kinetics.** Time-resolved assays (and PMSF titration) to separate self-cleavage from decarboxylation rates and test whether maturation is rate-limiting in vivo.
  5. **Test bypass generality.** Screen genomes flagged for Cls-family enzymes and experimentally test whether *psd* deletions are rescuable by exogenous ethanolamine beyond *Xanthomonas*.
  6. **Separate physical vs. specific PE roles.** Systematic complementation of PE-less *E. coli* with a graded panel of non-bilayer surrogate lipids while assaying specific PE-dependent functions (e.g., LacY folding, division) to define which phenotypes require PE per se.
- 

*Uncertainty statement:* Structural mechanism is strongest for the *E. coli* Type I PssA and *E. coli* Psd. Type II PssA mechanism, in-vivo regulation of PssA membrane association, and the precise self-cleavage trajectory of Psd remain partly inferential. Essentiality and downstream tailoring vary by lineage and growth condition and should not be generalized from *E. coli*.

---